

Blown-Fuse Indicator for AC Load

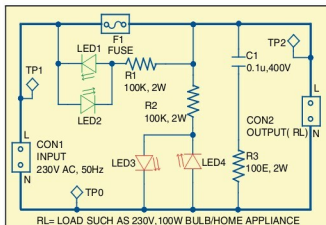


Fig. 1: Circuit of blown-fuse indicator for AC load

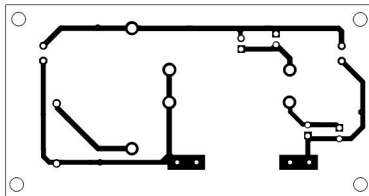


Fig. 2: An actual-size, single-side PCB for the blown-fuse indicator

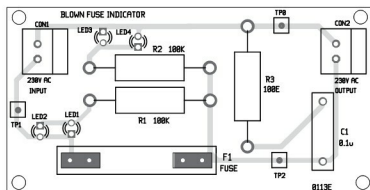


Fig. 3: Component layout for the PCB

TABLE I
LED Status Corresponding to Mains Supply and Fuse State

Mains supply	State of the fuse	Load	Green LEDs	Red LEDs
Present	Ok	Connected	Off	On
Present	Not Ok	Connected	On	Off
Present	Ok	Not connected	Off	On
Present	Not Ok	Not connected	On	On
Absent	NA	NA	Off	Off

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This simple circuit to monitor the state of fuse and mains power supply is highly useful for AC-powered appliances. Sometimes these appliances suddenly stop working and we don't have any clue what has gone wrong. The problem can be so simple that it can be rectified just by replacing the fuse. This circuit helps to identify such problems.

Circuit and working

The circuit is based on two resistors (R1 and R2) and four LEDs (two green and two red). LEDs provide the monitoring status as shown in Table I. R1 and R2 are current-limiting resistors for green and red LEDs, respectively. Lower-value resistors can be used in order to increase the intensity of LEDs.

The R-C network designed around C1 and R3 is used to filter the mains power supply, which is useful for most of the equipment. For proper functioning of the circuit, resistor values should be chosen as per the following relationship:

$$R_1 \ll R_2, R_2 \\ R_1 \sim R_2$$

TABLE II
Test Points

Test point	Details
TP0	Neutral
TP1	230V AC (Line voltage)
TP2	230V AC (If fuse ok)

PARTS LIST

Semiconductors:	
LED1, LED2	- 5mm green LED
LED3, LED4	- 5mm red LED
Resistors (all 2-watt, $\pm 5\%$ carbon):	
R1, R2	- 100-kilo-ohm, 2W
R3	- 10-ohm, 2W
Capacitors:	
C1	- 0.1 μ F, 400V polyester
Miscellaneous:	
CON1, CON2	- Two-pin connector
RL	- 230V, 100W bulb

R1 and R2 can have values ranging from 47 kilo-ohms to 100 kilo-ohms.

Table I show the LED status corresponding to the fuse and mains power supply state. If any of the LEDs glows, it means that mains power supply is connected. If green LEDs glow but red LEDs don't, the fuse is blown. If red LEDs glow but green LEDs don't, both the power supply and fuse are OK. If both red and green LEDs glow, this indicates that the fuse is blown and load disconnected.

Construction and testing

An actual-size, single-side PCB for blown-fuse indicator is shown in Fig. 2 and its component layout in Fig. 3. Assemble the circuit on a PCB and enclose in a small cabinet. Use a suitable fuse holder to connect the fuse on the PCB. Fix the LEDs on the front panel of the cabinet. For correct operation, check test-point voltages as per Table II. ●

The author has worked as a researcher and assistant professor in Technical University of Sofia (Bulgaria) and as lecturer and expert-lecturer in Kingdom of Morocco. Now he is an electronics engineer in the private sector in Bulgaria